18 - Carboxylic acids & derivatives

05/01/25

Production of carboxylic acids

1) further oxidation of aldehydes formed from primary alcohols

reagents : acidified $K_2Cr_2O_7$ / $KMnO_4$

how : reflux to form carbox. acid after distillation which formed the aldehydes

products : carboxylic acid

2) hydrolysis of nitriles

reagents : dilute acid or dilute alkali

how : heat, followed by acidification (if using alkali)

products : carboxylic acid & ammonium salt (with acid hydrolysis) or carboxylate salt (with alkaline hydrolysis)

* 3) hydrolysis of esters

reagents : dilute acid or dilute alkali

how : heat, followed by acidification (if using alkali)

products : carboxylic acid & alcohol

Reactions of carboxylic acids

- redox reaction with reactive metals

carboxylic acid + reactive metal $\longrightarrow$ salt + hydrogen gas

e.g $CH_3COOH + Mg \longrightarrow (CH_3COO)_2Mg + H_2$

- neutralisation reaction with alkalis

carboxylic acid + alkali $\longrightarrow$ salt + water

e.g $CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$

- acid-base reaction with carbonates

carboxylic acid + carbonate $\longrightarrow$ salt + water + carbon dioxide

e.g $2CH_3COOH + Na_2CO_3 \longrightarrow 2CH_3COONa + H_2O + CO_2$

- reduction by $LiAlH_4$

carboxylic acid + x[H] $\xrightarrow{LiAlH_4}$ primary alcohol

e.g $CH_3COOH + 4[H] \longrightarrow CH_3CH_2OH$

* • esterification with alcohols

carboxylic acid + alcohols $\xrightarrow[heat]{conc.\ H_2SO_4}$ ester + water

e.g $CH_3COOH + C_2H_5OH \longrightarrow CH_3COOC_2H_5 + H_2O$